

Victor Zhou

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EDUCATION

The University of Sydney

Bachelor of Engineering Honours (Aeronautical) / Bachelor of Science (Mathematics)

2020-Current

- Distinction WAM
- Majoring in Space Engineering
- Majoring in Mathematics

EXPERIENCE

NHMicro

Mechanical Engineering Intern

Sydney, Australia
2023.11-Current

USYD School of Aeronautical, Mechanical and Mechatronics Engineering

Technical Assistant

Sydney, Australia
2023.04-Current

Sydney Motorsports

Dynamics Department Head

Sydney, Australia
2021.03-2023.12

CUAVA

Undergraduate Engineering Intern

Sydney, Australia
2021.09-2022.01

USYD School of Aeronautical, Mechanical and Mechatronics Engineering

Undergraduate Engineering Intern

Sydney, Australia
2021.02-2021.02

TRANSFERABLE SKILLS

Leadership and Management

Leading the Dynamics Department of Sydney Motorsports where I oversee 30 student engineers directing the design of aerodynamics package, suspension systems and testing plans. Delegated tasks to the appropriate team members while ensuring newer members are adequately up skilled.

Problem Solving

Rapidly responded to critical on track failures on the Formula SAE car and ensuring it was ready and driving same day while only using limited track side tooling and equipment.

Communication

Actively manage documentation on a diverse range of machines within the School of AMME's FabLab Makerspace. Conduct frequent lab and machine inductions to ensure undergraduate, HDR and PhD students are adequately trained on FabLab equipment.

Time Management

Ensure that design deadlines are met for the Formula SAE car while studying full time and working part time as a technical officer. Organised weekly meetings for the three subsections under the Dynamics Department for our Formula SAE team.

TECHNICAL SKILLS

Computer Simulation

Fluent in writing automated journal files for testing an aerodynamics package over a range of yaw angles using Ansys Fluent. Conducted a topology optimisation study to develop a rear wing mounting solution 50% lighter than the previous years.

Programming (Python & MATLAB)

Fluent in writing analysis code in MATLAB and Python. Created various of orbital simulators for different problems using a range of both numerical integration schemes and analytically. Created various numerical solvers focusing on fluid flow including panel methods and vortex methods.

Composites Manufacturing:

Manufactured a full aerodynamics package for Sydney Motorsports using a combination of hand layup and prepreg techniques. The molds for these components were manufactured using a combination of free hand shaping, CNC hot wire cutting, CNC routing and 3D printing.

Computer Aided Design (SolidWorks & Fusion 360):

Designed multiple iterations of an aerodynamics package for Sydney Motorsports using SolidWorks. All CAD was made to be fully parameterised to make generating different configurations easy. Developed a fully parameterised wheel suitable to the British Standard 978. Created a fully featured switch assembly recreating a standard Cherry MX switch in SolidWorks. Ran several workshops focused on upskilling my university's Formula SAE team in CAD practices. Created parametric gears to variety of standards for use in mechanical wristwatches.

Machining

Machined replacement push rod mounts for Sydney Motorsports using a Pacifica FTV-2 manual mill. Machined interface plates for USYD Rocketry's hybrid engine testing cell. Proficient in the use of measuring tools such as bore gauges, vernier calipers.

Small Parts Handling and Finishing

Conducted final finishing on the main bridge and balance cock for the NH55 series of watches. Completed the polish on the chamfer on the going train wheels on the the NH55. Conducted metrology on parts with sub-millimetre sized features.

PROJECTS

Straight Line Watch Demonstrator

Designed a proof of concept straight line demonstrator to validate the gear train calculations for a custom, in-house manufactured watch movement.

Semi-Automated CFD Workflow

Improved the robustness of the previous semi-automated CFD workflow of Sydney Motorsports by creating a standard geometry template. Given a geometry and a set of yaw angle and velocity cases, the working scripts will generate a journal file and an intermediary CAD file for each case. These intermediary files are then submitted to Artemis, a HPC cluster for solving. Ansys FLUENT 2020R1 was used for meshing and solving.

Wind Tunnel Wing Section

Designed and manufactured a small scale wind tunnel model of the Selig-1223 profile. Used various manufacturing techniques such as conventional FDM printing as well as resin casting to allow for pressure tapping along the entire chord of the aerofoil even in the 1.5mm thick trailing edge sections. Undertook wind tunnel test with the model and verified the numerical results from xFoil and Ansys FLUENT.

Aerodynamics Package for FSAE Car

Planned and undertook the manufacture of the first in house aerodynamics package for Sydney Motorsport's. Utilised a number of different low cost manufacturing processes from FDM printing to CNC hotwire cutting to manufacture tooling. Created carbon fibre wing elements and end plates using a combination of hand lay-up techniques as well as using pre-impregnated carbon fibre.

Keyboard Design and Manufacture

Designed several custom keyboard cases for accepting standard DZ60 keyboard PCBs. Commissioned to design and manufacture a Pokemon styled macropad. Designed and manufactured a split-keyboard based off the Corne split-keyboard. Hand soldered and lubricated switches to a Pro-Micro micro controller using jumper wires. Designing and prototyping a set of custom spherical keycap profiles.